MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 Admission Onwards)

Value Added Course – MG3VACBCH200- Microplastics and Environment

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A

Multiple Choice Questions

Answer all questions

Each question carries 1 mark

1. What is the typical size range of microplastics? [K][1]
   A). 0.1 mm – 10 mm B). 1 nm – 5 mm C) 10 mm – 100 mm D) 0.001 mm – 0.01 mm

2. How do microplastics contribute directly to climate change? [U][4]
   A). By reflecting solar radiation B). By increasing ocean salinity C). By emitting greenhouse gases D). By reducing atmospheric oxygen

3. Which of the following best describes bioaccumulation? [K][3]
   A) Movement of microplastics through ecosystems B) Degradation of plastics into micro and nanoplastics C) Gradual buildup of microplastics in an organism's tissues D) Increase of microplastic levels in the ocean only

4. Which form of atmospheric deposition involves microplastics being removed from the air by rain or snow? [U][1]
   A) Dry deposition B) Condensation C) Wet deposition D) Sedimentation

5. Which of the following is NOT a likely result of microplastic contamination in agricultural soil? [K][2]
   A). Improved crop yield B). Reduced microbial diversity C). Altered soil structure D).Toxicity to plant roots

(5 x 1 =5)

Part B

Fill in the blanks

Answer all questions

Each question carries 2 marks

6. Plastics that result from the degradation of plastic items of size less than 1 nm are known as _____ plastics. [K][1]

7. Wastewater treatment plants cannot remove all microplastics, especially those that are _____or_____.. [U][2]

8. The process where microplastic concentration increases up the food chain is called _____. [U][2]

9. The main sources of microplastics in drinking water are_____and_____ [K] [3]

10. Biological method s microplastics removal are _____and_____ [U][4]

(5 x 2 =10)

Part C

Short answer questions

Answer any 5 questions

Each question carries 3 marks

11. Explain the classification of microplastics based on origin, size, and shape, with relevant examples. [U][1]

12. Differentiate between primary and secondary microplastics with one example each. [U][1]

13. Differentiate between bioaccumulation and biomagnification? [K][2]

14. What is atmospheric deposition, and how does it relate to transport of microplastics? [E][2]

15. Describe three physiological effects microplastic ingestion can have on marine organisms like fish or seabirds. [U][3]

16. How do microplastics alter marine food webs? Give two specific examples. [A][3]

17. What are three major ways microplastics contribute to climate change? [E][4]

(5 x 3 =15)

Part D

Short essay type questions

Answer any 4 questions

Each question carries 5 marks

18. Discuss the major environmental pathways of microplastics—air, water, and soil—and their associated impacts. [K][1]

19. How does wind contribute to the global distribution of microplastics? [E][2]

20. Analyse the primary sources of microplastics in food and drinking water, and discuss the current regulatory perspectives aimed at addressing this issue. [A][3]

21. Discuss how microplastics contribute to climate change through their effects on oceans, polar regions, and microbial ecosystems. [A][4]

22. Describe and compare the main strategies used for microplastic removal. [U][4]

23. Discuss the impact of microplastics on different human physiological systems, highlighting the associated health risks. [U][3]

(4x5=20)